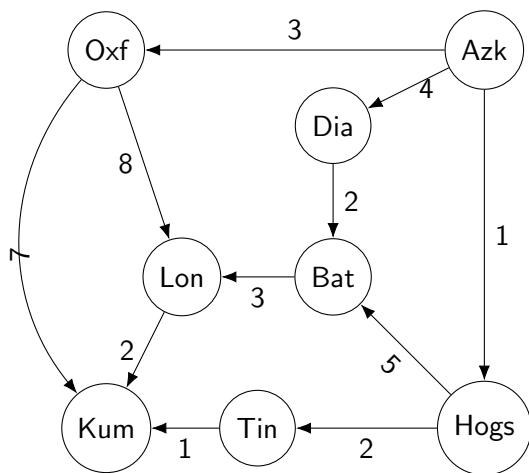


# CSE422

## Assignment 1

### Question 1

Consider the following weighted, directed graph.



**Heuristic Table**

| Node | $h(n)$ |
|------|--------|
| Azk  | 7      |
| Hogs | 6      |
| Dia  | 5      |
| Oxf  | 6      |
| Tin  | 2      |
| Bat  | 3      |
| Lon  | 2      |
| Kum  | 0      |

**Nodes:** Azkaban (Azk), Hogsmeade (Hogs), Diagon Alley (Dia), Oxford (Oxf), Tinworth (Tin), Bath (Bat), London (Lon), Kumara (Kum)

**Start:** Azkaban (Azk)    **Goal:** Kumara (Kum)

- Run A\* search from Azk to Kum. Show node expansions step by step.
- Is the path found by A\* optimal? Justify using heuristic properties.
- Check if all heuristic values satisfy the consistency condition. Give examples.
- Run Best First Search from Azk to Kum. Compare it with the A\* path and explain differences.
- Change heuristic of Hogs to 8. Re-run A\* and discuss whether the new heuristic affects optimality.

## Question 2

You are given a  $5 \times 5$  grid map. The agent starts at the bottom-left corner  $(4, 0)$  and must reach the top-right corner  $(0, 4)$ , avoiding obstacles.

Valid movements:

- Horizontal/Vertical (up, down, left, right): cost = 1
- Diagonal (e.g., top-right): cost =  $\sqrt{2}$

Heuristics:

$h_1(n)$  : Manhattan Distance to goal

$h_2(n)$  : Euclidean Distance to goal

- (a) Are both heuristics admissible? Explain.
- (b) If  $h_1(n)$  is multiplied by 1.6 in A\*, will the result be optimal?

## Question 3

- Design a weighted graph with at least 8 nodes.
- Define start and goal nodes.
- Create an admissible and consistent heuristic table.
- Apply an informed search algorithm to find the optimal path.

*Note: Your graph and values must be unique.*